



PHOTOSYNTHESIS — FULL MIND MAP

Biology Ch. 8.i · Plant Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Chlorophyll — Composition, Function & Chloroplast Self-Replication
2. Light Reaction vs Dark Reaction — Sites in Chloroplast
3. Raw Materials, Products, Oxygen Source & Glucose
4. Where & When Photosynthesis Occurs
5. Light Spectrum — Red, Blue, Violet & Green Light
6. Energy Conversion, ATP & Calvin Cycle
7. Photosynthetic Efficiency — Chlorella
8. Roots & Photosynthesis
9. Photosynthesis in Carbon Cycle & Night Respiration
10. Infrared Photosynthesis — Green Sulphur Bacteria

1. Chlorophyll — Composition, Function & Chloroplast Self-Replication

- Chlorophyll ke components mein carbon, hydrogen, oxygen, nitrogen aur magnesium shamil hain; calcium component nahi hai. Magnesium chlorophyll ke porphyrin ring ke centre mein present hota hai. [Q1, Q2, Q3]
- Photosynthesis mein chlorophyll ka main kaam light ko absorb karna hai. Source ke mutabik chlorophyll chloroplast mein located hota hai; absorbed light energy food synthesis ke liye use hoti hai aur green wavelength mainly reflect hone se plant green dikhta hai. [Q4]
- Chloroplast mein DNA aur RNA dono ki presence use self-replication ke liye qualified/semi-autonomous organelle banati hai. Source RNA primers ko DNA synthesis initiate karne mein important batata hai. [Q5]

EXAM TRAP — Chlorophyll Composition

- ▶ Magnesium = chlorophyll ka key element; Calcium = chlorophyll ka component NAHI. [Q1–Q3]
- ▶ Green light mainly reflect hoti hai; isi wajah se chlorophyll/plant green dikhta hai. [Q4]

2. Light Reaction vs Dark Reaction — Sites in Chloroplast

- Photosynthesis ke light-dependent reactions thylakoid/grana membranes mein hote hain aur ATP + NADPH produce karte hain. Light-independent/dark reaction (Calvin cycle/Blackman reaction) chloroplast ke stroma mein hota hai aur light reaction se bane ATP/NADPH ka use karke CO₂ ko glucose mein convert karta hai. [Q6]
- Solar energy ka ATP mein conversion chloroplast ke light reactions mein hota hai; source ATP aur NADPH ko energy-transporting molecules ke roop mein describe karta hai. [Q29]

3. Raw Materials, Products, Oxygen Source & Glucose

- Photosynthesis ka main product glucose (sugar) hai aur oxygen (O₂) mainly by-product ke roop mein release hota hai. Glucose ka chemical formula C₆H₁₂O₆ hai. [Q7, Q10]
- Photosynthesis mein release hone wala oxygen carbon dioxide se nahi, water se aata hai. Light reaction mein water molecules ki photolysis/splitting se oxygen evolve hota hai. [Q8, Q15]
- Green plants photosynthesis mein atmosphere se CO₂ lete hain aur O₂ release karte hain while glucose banta hai. Carbon dioxide essential raw material hai; sunlight aur water bhi required hain, jabki oxygen process ke liye required nahi hai. [Q9, Q11, Q12, Q13, Q14]

EXAM TRAP — CO₂ vs O₂

- ▶ CO₂ = raw material/absorbed gas; O₂ = by-product/released gas. [Q9, Q11–Q14]
- ▶ Released O₂ ka source = H₂O ki photolysis, CO₂ nahi. [Q8, Q15]

4. Where & When Photosynthesis Occurs

- Source ke according Earth par approximately 90% photosynthesis aquatic plants aur algae se hota hai; isme 85% sea mein (mainly phytoplankton/algae) aur remaining 5% rivers/ponds etc. mein bataya gaya hai. [Q16]
- Photosynthesis light ki presence mein hota hai. Source normally sunlight ko main source batata hai, marine algae ke liye moonlight ka mention karta hai aur electric/artificial light mein bhi photosynthesis possible batata hai; is framing ke hisab se suitable light ho to day ya night dono mein ho sakta hai. [Q17, Q26]
- Plants food banane ki process ko Photosynthesis kehte hain. Ye primarily leaves mein aur partially stems mein hota hai; cellular site chloroplast hai. Broad leaf lamina sunlight aur CO₂ absorption mein help karti hai. [Q18, Q19, Q20]

5. Light Spectrum — Red, Blue, Violet & Green Light

- Chlorophyll mainly blue, red aur violet light absorb karta hai. Source red + blue regions ko strongest photosynthetic wavelengths batata hai; green light mainly reflect hoti hai aur photosynthesis green mein very low/absent bataya gaya hai. [Q21, Q22]
- Source visible spectrum ko 3800–7600 Å range mein place karta hai. Best photosynthesis red light mein, next best blue light mein aur green light mein nil bataya hai; T.W. Engelmann (1888) ke action-spectrum experiment ka bhi mention hai. [Q23]
- Given exam questions ke according red light/rays photosynthesis ke liye most effective/most utilized hain. [Q24, Q25]

EXAM TRAP — Light Colour Order

- ▶ Most effective/active = RED; next = BLUE; GREEN = mainly reflected / source mein negligible-to-nil photosynthesis. [Q21–Q25]

6. Energy Conversion, ATP & Calvin Cycle

- Photosynthesis light energy ko chemical energy mein convert karta hai. Source equation: $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$. [Q27]
- Dark reaction/Calvin cycle mein one glucose synthesize karne ke liye source 6 turns batata hai; each turn 3 ATP + 2 NADPH use karta hai, total = 18 ATP + 12 NADPH. [Q28]
- Light energy ka chemical energy mein conversion photosynthesis se hota hai; sunlight energy green plants mein chemical energy mein transform hoti hai. [Q30, Q31]
- Photosynthesis mein solar energy molecular bonds mein chemical energy ke roop mein store hoti hai, jise source potential energy classify karta hai. Sunlight ki free energy ko usable chemical potential energy mein badalne ke karan photosynthesis biological systems ke metabolic energy ka ultimate source bataya gaya hai. [Q32, Q33]

7. Photosynthetic Efficiency — Chlorella

- Given options mein Chlorella ko solar energy ko chemical energy mein convert karne ki maximum efficiency wala organism marked kiya gaya hai. Source Chlorella ko single-celled green alga (Chlorophyceae) batata hai aur photosynthetic efficiency up to 8% mention karta hai. [Q34]

8. Roots & Photosynthesis

- Generally roots ka kaam photosynthesis nahi hai. Source ke mutabik root cells normally chloroplasts nahi rakhte; roots water/nutrients absorb karte hain aur plant ko anchor karte hain, jabki food preparation mainly leaves mein hoti hai. [Q35]

9. Photosynthesis in Carbon Cycle & Night Respiration

- Carbon cycle mein photosynthesis atmospheric CO_2 ko consume karke uski amount decrease karta hai; respiration, organic matter decay aur volcanic action CO_2 release karke amount increase karte hain. [Q36]
- Source ke night-tree questions ke framing mein sunlight na hone par photosynthesis/oxygen production nahi hota, while trees respiration continue karke CO_2 release karte hain; isi basis par raat mein bade ped ke neeche sone ko harmful/not advised bataya gaya hai. [Q37, Q38]



SOURCE-SPECIFIC NOTE — Night Under Trees

- ▶ Q37–Q38 ka statement/explanation source ke exam-oriented framing ke according preserve kiya gaya hai: night respiration → CO_2 release; photosynthesis absent → O_2 production absent. [Q37, Q38]

10. Infrared Photosynthesis — Green Sulphur Bacteria

- Source ke according deep-sea green sulphur bacteria sunlight ke invisible infrared radiations ka use photosynthesis ke liye kar sakte hain. Ye water ko reducing agent ke roop mein use nahi karte aur oxygen produce nahi karte; isse anoxygenic photosynthesis kaha gaya hai. [Q39]